

FRANKSTON, Australia

WMO: 948710

Lat: 38.1481S

Lon: 145.1156E

Elev: 6

StdP: 101.25

Time Zone: 10.00 (AUS)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	6.7	1.5	4.2	11.0	2.5	4.5	10.4	16.6	11.5	14.1	11.1	2.1	110	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.7	31.1	19.0	29.0	18.9	27.1	18.4	21.1	26.7	20.2	25.9	19.4	24.9	3.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.3	22.8	18.4	13.3	22.4	17.4	12.4	21.9	61.1	26.8	58.1	25.9	55.4	25.1	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.7	11.9	10.5	DB	3.5	36.1	0.9	2.4	2.9	37.9	2.4	39.3	1.9	40.6	1.3	42.3
			WB	2.9	22.7	0.9	1.0	2.2	23.4	1.7	24.1	1.2	24.6	0.6	25.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	19.6	19.9	18.4	15.5	13.4	11.2	10.5	11.1	12.6	14.0	16.1	17.8	(6)
	DBStd	4.22	3.47	3.31	3.06	2.48	2.07	1.83	1.61	1.85	2.27	2.78	3.39	3.40	(7)
	HDD10.0	32	0	0	0	0	1	8	13	8	2	1	0	0	(8)
	HDD18.3	1458	24	18	38	92	154	214	243	226	174	139	85	52	(9)
	CDD10.0	1847	298	278	259	164	104	44	28	41	78	124	184	243	(10)
	CDD18.3	232	63	63	39	6	0	0	0	0	1	4	19	37	(11)
	CDH23.3	1609	477	413	234	18	0	0	0	0	2	22	141	302	(12)
	CDH26.7	507	179	131	59	2	0	0	0	0	0	2	33	99	(13)
(14) Wind	WSAvg	4.4	4.2	3.8	3.8	3.8	4.4	4.3	4.9	4.8	4.9	4.8	4.4	4.4	(14)
(15) Precipitation	PrecAvg	736	48	53	51	71	59	70	63	63	72	65	67	56	(15)
	PrecMax	1000	132	161	85	180	122	142	129	108	151	134	143	138	(16)
	PrecMin	491	4	7	28	24	17	18	18	16	26	13	22	0	(17)
	PrecStd	143	31	39	16	37	29	31	26	24	32	33	31	33	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	32.9	31.0	25.4	20.6	17.7	15.9	19.0	22.8	26.0	30.3	33.0	(19)
		MCWB	20.3	19.1	18.6	16.5	14.5	13.5	11.3	12.8	14.4	15.9	17.9	18.4	(20)
	2%	DB	31.1	30.3	28.2	23.1	19.0	16.0	14.6	17.0	20.0	23.1	27.0	29.2	(21)
		MCWB	19.5	19.3	18.6	15.8	14.0	12.4	10.9	12.0	13.8	15.4	17.5	18.1	(22)
	5%	DB	28.4	28.0	25.9	21.4	18.0	14.8	13.7	15.3	18.3	21.1	24.6	26.7	(23)
		MCWB	19.1	18.9	18.1	15.5	13.7	11.7	10.4	11.3	12.9	14.4	16.8	17.9	(24)
	10%	DB	25.8	25.8	23.7	19.7	16.8	14.0	13.0	14.1	16.6	19.1	22.2	24.1	(25)
		MCWB	18.9	18.7	17.5	15.0	13.3	11.3	10.2	10.6	12.2	13.7	16.1	17.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.1	21.1	18.0	16.1	14.4	12.6	14.0	16.1	17.5	20.1	20.9	(27)
		MCDB	27.5	27.2	26.9	22.3	18.1	16.3	14.3	17.3	20.1	22.5	25.8	26.8	(28)
	2%	WB	21.3	21.0	19.9	17.0	15.0	13.0	11.6	12.6	14.5	16.2	18.6	19.7	(29)
		MCDB	27.1	25.8	25.0	20.3	17.6	15.0	13.5	16.0	18.9	21.1	24.1	25.9	(30)
	5%	WB	20.2	20.2	18.9	16.3	14.3	12.3	11.0	11.6	13.4	15.2	17.6	18.7	(31)
		MCDB	26.4	25.3	23.7	19.8	17.0	14.2	12.9	14.5	17.2	20.0	23.0	24.7	(32)
	10%	WB	19.2	19.3	18.0	15.6	13.6	11.7	10.6	11.0	12.5	14.1	16.6	17.7	(33)
		MCDB	25.0	24.4	22.8	19.0	16.3	13.6	12.4	13.3	15.8	18.4	21.6	23.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	7.7	7.0	5.7	4.5	4.1	3.8	4.3	5.2	6.3	7.2	7.8	(35)
		MCDBR	13.7	12.7	11.5	8.8	6.3	5.0	4.9	6.6	8.2	10.9	12.7	13.8	(36)
	5% WB	MCWBR	5.0	4.8	4.6	4.2	3.8	3.2	3.1	3.7	4.4	5.1	5.3	5.4	(37)
		MCWBR	10.9	9.9	9.1	7.1	5.1	4.2	3.7	5.5	7.3	9.2	10.2	11.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.357	0.359	0.336	0.332	0.319	0.306	0.309	0.321	0.342	0.343	0.350	0.346	(40)
		taud	2.503	2.508	2.575	2.556	2.575	2.599	2.580	2.540	2.476	2.492	2.500	2.529	(41)
	Ebn at Noon		975	949	933	871	823	806	825	871	909	954	976	991	(42)
		Edh at Noon	112	107	92	82	70	63	68	82	100	108	112	111	(43)
(44) All-Sky Solar Radiation	RadAvg		7.08	6.19	4.90	3.40	2.25	1.76	1.98	2.79	4.00	5.36	6.30	7.04	(44)
	RadStd		0.43	0.31	0.27	0.19	0.12	0.14	0.10	0.17	0.26	0.40	0.35	0.40	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-84	+141	N/A	(46)
(47)	Regional (36 neighbors)	+0.51	N/A	N/A	+0.84	N/A	N/A	-18	-130	+174	+58	(47)

Nomenclature: See separate page